

AdaptiveBargainNegotiator: A Hybrid Agent Submitted to the ANAC 2026 HAN League

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Abstract

AdaptiveBargainNegotiator is a hybrid negotiation agent for the ANAC 2026 Human-Agent Negotiation (HAN) league. The agent separates formal negotiation decisions from natural-language communication. Offers and acceptance decisions are produced by a BOA-style negotiation core using a time-dependent offering strategy, a frequency-based opponent model, and an adaptive acceptance policy. Natural-language messages are generated by a large language model (LLM), but the LLM is not allowed to change the formal action selected by the negotiation core. This design is intended to keep the agent reliable under strict competition conditions while still providing cooperative and human-readable explanations of its behavior.

1 Introduction

The HAN league evaluates agents in bilateral negotiations in which formal offers are accompanied by free-form textual communication. This setting differs from purely numerical automated negotiation because the agent should not only obtain good utility, but also communicate in a way that is understandable, cooperative, and persuasive to a human or LLM-based counterpart.

AdaptiveBargainNegotiator was designed around this distinction. The agent uses a deterministic negotiation mechanism for the critical parts of the protocol: choosing offers, deciding whether to accept, and respecting utility constraints. An LLM is used as a communication layer that explains these choices in concise natural language. This reduces the risk of malformed actions or utility calculation errors caused by relying on the LLM for the whole negotiation.

2 The Design of AdaptiveBargainNegotiator

2.1 Negotiation Choices

The formal negotiation strategy follows the BOA architecture: a bidding component, an opponent model, and an acceptance component. The bidding component uses a time-dependent concession curve as its main structure. Early in the negotiation, the agent targets high-utility offers. As the deadline approaches, the target utility gradually decreases to improve the chance of reaching an agreement.

The agent also uses a frequency-based opponent model. Among offers that satisfy the agent's current utility target, it prefers offers estimated to be more acceptable to the opponent. This creates a compromise between protecting the agent's own utility and proposing outcomes that are likely to be accepted.

The concession behavior is adaptive. If the opponent appears to be conceding toward the agent, the agent avoids conceding too quickly. If the opponent appears rigid after a significant part of the negotiation has passed, the target utility is modestly relaxed. This keeps the agent from being purely conceder-like while still allowing it to respond to hard opponents.

The acceptance component extends an ACNext-style policy with a dynamic utility threshold. The agent compares the opponent's current offer with a deadline-aware threshold and with the

utility it expects from its next own proposal. Near the end of the negotiation, the threshold becomes more flexible, especially against opponents that show little concession.

2.2 Concurrent Negotiation

The HAN setting considered here is a bilateral negotiation session rather than a simultaneous multi-market environment. AdaptiveBargainNegotiator therefore does not maintain a portfolio of concurrent negotiations or transfer commitments between sessions.

Instead, each negotiation is treated as an independent session. The agent stores only within-session information such as the sequence of opponent offers and the current opponent-model estimates. This avoids leakage of information across matches and makes the behavior reproducible when the agent is evaluated on different scenarios or opponents.

2.3 Natural-Language Communication

The agent uses an LLM to generate short persuasive messages for the selected formal action. The prompt instructs the LLM to explain the agent’s move in a professional and cooperative tone while preserving the agent’s utility. The LLM is asked to return a structured JSON object containing only the message text.

Importantly, the LLM does not decide which offer to send or whether to accept an offer. This separation is useful in HAN because natural communication matters, but formal decisions must remain valid, consistent, and utility-aware. The LLM layer can improve perceived cooperativeness without compromising the safety of the negotiation core.

2.4 Risk Management

The main risks in LLM-based negotiation agents are invalid action formats, slow model responses, hallucinated utility reasoning, and inconsistent concession behavior. AdaptiveBargainNegotiator mitigates these risks in several ways.

First, formal decisions are produced by the rule-based BOA core rather than by the LLM. Second, the LLM output is limited to the textual message and requested in a simple JSON format. Third, if the LLM backend fails or times out, the agent falls back to short deterministic messages instead of crashing. This fallback is only a communication fallback; the negotiation action itself is still generated by the same formal strategy.

The agent also protects against poor agreements by rejecting offers below its reservation value and by using a dynamic acceptance threshold. These mechanisms aim to reduce the chance of accepting low-utility offers early while still allowing agreement when the deadline is close.

3 Evaluation

We evaluated the agent locally using the HAN 2026 template environment. Unit tests from the template passed, and the agent was able to complete sample negotiations against the provided template-based opponent. In a short Grocery scenario run against the Template-BasedAdapterNegotiator, the agent reached an agreement and obtained higher individual advantage than the opponent.

The local tests mainly verified implementation correctness, dependency compatibility, and basic behavior of the BOA-plus-LLM architecture. They should not be interpreted as a complete performance guarantee. HAN performance can depend strongly on the scenario, the human or LLM opponent, latency of the LLM backend, and the subjective evaluation of the generated messages.

4 Lessons and Suggestions

The most important design lesson is that reliability and communication should be handled separately. Letting an LLM decide the whole action is attractive, but it creates avoidable risks in utility calculation, output formatting, and consistent concession. A more conservative design is to use a negotiation algorithm for decisions and an LLM for explanation.

Another lesson is that opponent modeling must influence actual strategic choices. The current agent uses a frequency-based model to choose more acceptable offers within a target utility range, but further improvements are possible. Future versions could add stronger opponent-type detection, explicit prediction of future opponent offers, and more careful reuse of the opponent’s best previous offer near the deadline.

Conclusions

AdaptiveBargainNegotiator is a hybrid HAN agent that combines a robust BOA-style negotiation core with LLM-generated natural-language communication. The design prioritizes valid actions, utility-aware concession, adaptive acceptance, and concise cooperative messages. This makes it a practical baseline for HAN, while leaving clear opportunities for improving opponent modeling and late-stage agreement behavior in future versions.